

Frobenius splitting and singular loci of prime-size permanents

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Abstract

For each prime p , we compute a Frobenius trace coefficient associated with the maximal permanents of a generic $(p-1) \times p$ matrix. A cyclic permutation of the columns makes this coefficient equal to one in characteristic p . We prove that the maximal permanents form a radical complete intersection of height p in characteristic p and in characteristic zero. A local differential argument bounds every Jacobian-rank stratum and gives ambient codimension exactly $2p$ for the singular locus of the $p \times p$ permanent in both characteristics. A dimension estimate obtained by fixing a nonzero border entry propagates the bound to arbitrary matrix sizes. In characteristic zero, if q is the largest prime at most n , the singular codimension lies between $2q$ and $2n$, and hence is asymptotic to $2n$. We also obtain exact critical codimensions for partial row expansions and exact strength at prime sizes.

1 Introduction and main results

Let $X = (x_{ij})$ be a generic $n \times n$ matrix over a field k . Its permanent is

$$\text{per}_n(X) = \sum_{\sigma \in S_n} \prod_{i=1}^n x_{i, \sigma(i)}.$$

The first partial derivatives of per_n are its permanental cofactors. Thus the geometry of the singular locus of the permanent is closely related to the codimension of rectangular permanental ideals.

For an $r \times n$ generic matrix A , with $r \leq n$, let $P_r(A)$ denote the ideal of its maximal permanents. Boralevi, Carlini, Michałek, and Ventura conjecture that the maximal permanents of a generic $r \times (r+1)$ matrix form a complete intersection [1, Conjecture 3.4]. Their work establishes several small cases and gives a uniform lower bound of six for the ambient singular codimension of per_n when $n \geq 6$ [1, Theorem 4.23]. We treat all row sizes one less than a prime, and obtain asymptotically sharp bounds for square permanents of arbitrary size.

Throughout, dimensions are geometric dimensions: they may be computed after extension to an algebraic closure. The critical locus of a polynomial is the common zero set of all its first derivatives. For the permanent it lies on the hypersurface, in every characteristic, by expansion along a row. We write

$$s_k(n) = \text{codim}_{\mathbb{A}_k^{n^2}} \text{Crit}(\text{per}_n), \quad c_k(r, n) = \text{ht } P_r(A).$$

All codimensions in this paper are measured in matrix space, rather than inside the permanental hypersurface.

Theorem 1.1. *Let p be prime, and let A be a generic $(p-1) \times p$ matrix. Over every field of characteristic p or zero, the ideal $P_{p-1}(A)$ is a geometrically reduced complete intersection of height p . Its Hilbert series is*

$$\frac{(1 - z^{p-1})^p}{(1 - z)^{p(p-1)}}. \quad (1.1)$$

In characteristic p , its quotient is Frobenius split. In these same two characteristics,

$$s_k(p) = 2p. \quad (1.2)$$

The splitting in this theorem uses the prime equal to the number of columns. The assertion is not a statement of splitting in every positive characteristic for a fixed matrix size. Characteristic dependence of permanent F -singularities is studied in [2].

Theorem 1.2. *Suppose k has characteristic zero and $n \geq 2$. If q is the largest prime at most n , then*

$$2q \leq s_k(n) \leq 2n, \quad q \leq c_k(n-1, n) \leq n. \quad (1.3)$$

Consequently,

$$\lim_{n \rightarrow \infty} \frac{s_k(n)}{2n} = 1, \quad \lim_{n \rightarrow \infty} \frac{c_k(n-1, n)}{n} = 1.$$

The proof begins with a coefficient identity. If f_j is the maximal permanent of A omitting column j and $H = \prod_j f_j$, then

$$\left[\prod_{i=1}^{p-1} \prod_{j=1}^p a_{ij}^{p-1} \right] H^{p-1} = 1 \quad \text{in } \mathbb{F}_p. \quad (1.4)$$

This is a specific instance of the relation between finite-field point counting and Frobenius splitting developed by Knutson [3]. We give two direct cyclic proofs. We then derive a local rank estimate from the nonzero coefficient, without assuming a complete intersection in advance. This separates the dimension argument from the later splitting and reducedness assertions.

Theorem 1.3. *Use the maximal permanents $f_1, \dots, f_p$ above, and let $S \subseteq \{1, \dots, p\}$ have cardinality $s \geq 1$. Over a field of characteristic p or zero, put*

$$g_S(A, u) = \sum_{j \in S} u_j f_j(A).$$

In the affine space with coordinates A and $(u_j)_{j \in S}$,

$$\text{codim Crit}(g_S) = 2s.$$

Its strength, counted as the minimum number of products of positive-degree homogeneous polynomials, is exactly s .

The strength consequence for arbitrary square size is

$$q \leq \text{str}(\text{per}_n) \leq n \quad (\text{char } k = 0). \quad (1.5)$$

This convention for strength counts products, with no subtraction of one. Neither this invariant nor the singular codimension alone establishes an unrestricted arithmetic-circuit lower bound.

2 The cyclic trace coefficient

Let $N = p(p-1)$, let $A = (a_{ij})$ be a generic $(p-1) \times p$ matrix, and set

$$f_j = \text{per}(A_{\widehat{j}}), \quad H = \prod_{j=1}^p f_j, \quad M = \prod_{i,j} a_{ij}.$$

Both H and M have degree N .

Proposition 2.1. *For every prime p , the integer coefficient $[M^{p-1}]H^{p-1}$ is congruent to one modulo p .*

Proof. For a nonnegative integer e ,

$$\sum_{a \in \mathbb{F}_p} a^e = \begin{cases} -1, & e > 0 \text{ and } p-1 \mid e, \\ 0, & \text{otherwise.} \end{cases}$$

In a monomial of total degree $N(p-1)$, a nonzero sum over $\mathbb{F}_p^N$ requires every exponent to be a positive multiple of $p-1$. Thus each exponent must be exactly $p-1$. Since N is even, we obtain

$$[M^{p-1}]H^{p-1} = \sum_{A \in \text{Mat}_{p-1,p}(\mathbb{F}_p)} H(A)^{p-1} \text{ in } \mathbb{F}_p. \quad (2.1)$$

Cyclic permutation of the p columns preserves the summand. Every orbit has size one or p , so only fixed matrices contribute modulo p . A fixed matrix has all columns equal to a vector $a = (a_1, \dots, a_{p-1})^t$. Its maximal permanents all equal $(p-1)! \prod_i a_i$. Hence its contribution is one exactly when every a_i is nonzero, and is zero otherwise. The sum is therefore $(p-1)^{p-1} = 1$ in $\mathbb{F}_p$. The argument also applies when $p = 2$. $\square$

There is also a coefficient-level proof, which makes the expansion multiplicities explicit. It will be useful as an independent check on the point-counting argument.

A second proof of Proposition 2.1. Index columns by $\mathbb{F}_p$ and label the $p-1$ occurrences of each f_j by $h \in \{1, \dots, p-1\}$. A term in the product is a family of bijections

$$\sigma_{j,h} : \{1, \dots, p-1\} \longrightarrow \mathbb{F}_p \setminus \{j\}.$$

It contributes to the required coefficient if, for each row i and column a , exactly $p-1$ pairs (j, h) satisfy $\sigma_{j,h}(i) = a$. Call this the balance condition. The transformation

$$(T\sigma)_{j,h}(i) = \sigma_{j-1,h}(i) + 1$$

preserves balance and has order p . A fixed family is precisely one of the form $\sigma_{j,h}(i) = \tau_h(i) + j$, where each τ_h is a bijection from $\{1, \dots, p-1\}$ to $\mathbb{F}_p^*$. Every such family is balanced, since varying j visits each column once for fixed i, h . There are $((p-1)!)^{p-1}$ fixed families. Wilson's congruence shows that this count is one modulo p , including the immediate case $p = 2$. All other orbits have size p . $\square$

3 A local consequence of a nonzero coefficient

The next statements apply to a general product of homogeneous polynomials whose degrees sum to the number of variables.

3.1 Differential nonmembership

Let $R = k[x_1, \dots, x_N]$ in characteristic p , and let $f_1, \dots, f_t$ be positive-degree homogeneous polynomials with $\sum_i \deg f_i = N$. Put $H = \prod_i f_i$ and assume

$$c = [x_1^{p-1} \cdots x_N^{p-1}] H^{p-1} \neq 0. \quad (3.1)$$

For an ideal J , write $J^{[p]} = (a^p : a \in J)$.

Lemma 3.1. *For every prime $\mathfrak{p}$ of R , writing $\mathfrak{m} = \mathfrak{p}R_{\mathfrak{p}}$, one has $H^{p-1} \notin \mathfrak{m}^{[p]}$. The same statement holds for the product of any subset of the f_i , raised to the power $p-1$.*

Proof. Consider the composition of ordinary differential operators

$$D = \prod_{i=1}^N \left(\frac{\partial}{\partial x_i} \right)^{p-1}.$$

Any monomial with an exponent below $p-1$ is annihilated, and the total degree $N(p-1)$ leaves only the indicated monomial. Therefore

$$D(H^{p-1}) = ((p-1)!)^N c \in k^*. \quad (3.2)$$

Each derivation extends to $R_{\mathfrak{p}}$ and preserves $\mathfrak{m}^{[p]}$: for $a \in \mathfrak{m}$ and $b \in R_{\mathfrak{p}}$, $\partial(a^p b) = a^p \partial b$ in characteristic p . This also holds for fractions. Thus their composition preserves $\mathfrak{m}^{[p]}$, and (3.2) rules out membership of H^{p-1} . If a subproduct belonged to $\mathfrak{m}^{[p]}$, multiplying by the remaining factors would give the forbidden membership for H^{p-1} . $\square$

3.2 Two elementary local identities

Lemma 3.2. *If $(S, \mathfrak{m})$ is a regular local ring of dimension h and characteristic p , then*

$$\mathfrak{m}^{h(p-1)+1} \subseteq \mathfrak{m}^{[p]}. \quad (3.3)$$

Proof. The maximal ideal has h generators. In every monomial of degree $h(p-1)+1$ in those generators, one exponent is at least p . $\square$

Lemma 3.3. *Let S be a local ring of characteristic p , let $I = (h_1, \dots, h_s)$, and let $g = Uh$ for $U \in \mathrm{GL}_s(S)$. Then*

$$\prod_{i=1}^s g_i^{p-1} \equiv \det(U)^{p-1} \prod_{i=1}^s h_i^{p-1} \pmod{I^{[p]}}. \quad (3.4)$$

Proof. Work first in $S[y_1, \dots, y_s]/(y_1^p, \dots, y_s^p)$. Its top homogeneous component is free of rank one, generated by $\prod_i y_i^{p-1}$. A scaling $y_i \mapsto ay_i$ by a unit acts on this component by a^{p-1} . A permutation acts trivially, in agreement with its determinant to the power $p-1$, also when $p=2$. An elementary shear $y_i \mapsto y_i + ay_j$ fixes the component: every term other than y_i^{p-1} in the expansion has exponent at least p on y_j after multiplication by y_j^{p-1} . These operations generate the invertible matrices over a local ring, by elimination using a unit entry. They therefore prove the determinant character asserted in (3.4). Substitution $y_i = h_i$ gives the result. Entries of U need not be constants. $\square$

3.3 Height and Jacobian-rank strata

Assume now that k is perfect. Choose a subset of s of the polynomials in (3.1), relabel it $h_1, \dots, h_s$, and put $I_S = (h_1, \dots, h_s)$. Let J_h be their $s \times N$ Jacobian matrix. For $0 \leq r \leq s$, define the locally closed set

$$Z_r = \{x \in V(I_S) : \text{rank } J_h(x) = r\}.$$

Proposition 3.4. *Under (3.1), one has*

$$\text{ht } I_S = s, \quad \text{codim}_{\mathbb{A}^N} Z_r \geq 2s - r \quad (3.5)$$

for every nonempty rank stratum. The generators of I_S form a regular sequence.

Proof. At any prime $\mathfrak{p} \supseteq I_S$, set $S_0 = R_{\mathfrak{p}}$, $\mathfrak{m} = \mathfrak{p}S_0$, and $h = \text{ht } \mathfrak{p}$. If $h < s$, then

$$\prod_{i=1}^s h_i^{p-1} \in \mathfrak{m}^{s(p-1)} \subseteq \mathfrak{m}^{h(p-1)+1} \subseteq \mathfrak{m}^{[p]},$$

contradicting Lemma 3.1. The height theorem gives the opposite bound, so $\text{ht } I_S = s$. Cohen–Macaulayness of the polynomial ring gives the regular-sequence assertion.

For the refined estimate, take the generic point $\mathfrak{p}$ of the closure of an irreducible component of Z_r . Let $K = \text{Frac}(R/\mathfrak{p})$ and $h = \text{ht } \mathfrak{p}$. The span of the classes of the h_i in $\mathfrak{m}/\mathfrak{m}^2$ has dimension r . To justify this point, use the conormal sequence

$$\mathfrak{m}/\mathfrak{m}^2 \longrightarrow \Omega_{R/k} \otimes_R K \longrightarrow \Omega_{K/k} \longrightarrow 0. \quad (3.6)$$

The first space has dimension h because $R_{\mathfrak{p}}$ is regular; the middle one has dimension N . Since k is perfect and K/k is a finitely generated field extension, $\dim_K \Omega_{K/k} = \text{trdeg}_k K = N - h$. Exactness of (3.6) therefore makes its first map injective. Its values on the generator classes are exactly the Jacobian rows evaluated over K .

Row reduction over K , lifted to an invertible matrix over $R_{\mathfrak{p}}$, replaces the h_i by g_i with

$$g_1, \dots, g_r \in \mathfrak{m}, \quad g_{r+1}, \dots, g_s \in \mathfrak{m}^2.$$

Lemma 3.3, together with $I_S^{[p]} \subseteq \mathfrak{m}^{[p]}$, shows that $\prod_i g_i^{p-1} \notin \mathfrak{m}^{[p]}$. But this product belongs to $\mathfrak{m}^{(2s-r)(p-1)}$. If $h < 2s - r$, Lemma 3.2 would force membership in $\mathfrak{m}^{[p]}$, a contradiction. Thus $h \geq 2s - r$, as required. $\square$

The rank estimate did not use reducedness or a splitting on the quotient. Its only coefficient input was (3.1). We will use this separation when applying it to critical loci.

4 Compatible splitting and characteristic-zero transfer

4.1 The prime characteristic

Apply Proposition 2.1 to the maximal permanents $f_1, \dots, f_p$. Proposition 3.4 proves the height and regular-sequence statements over an algebraic closure of $\mathbb{F}_p$, and hence over every field of characteristic p by faithfully flat descent. We next establish reducedness and the claimed splitting.

Choose an additive coefficient map $\lambda : k \rightarrow k$ with $\lambda(a^p b) = a\lambda(b)$ and $\lambda(1) = 1$. It exists over any field: project k onto its 1-coordinate in a k^p -basis containing 1, and then take the p th root. For the N entries of A , define

$$T(bx^\alpha) = \begin{cases} \lambda(b)x^{(\alpha-(p-1)\mathbf{1})/p}, & \alpha_i \equiv p-1 \pmod{p} \text{ for all } i, \\ 0, & \text{otherwise.} \end{cases}$$

This is additive and satisfies $T(a^p g) = aT(g)$. Since H^{p-1} has degree $N(p-1)$, Proposition 2.1 implies $T(H^{p-1}) = 1$. Thus

$$\phi(g) = T(H^{p-1}g) \quad (4.1)$$

is a Frobenius splitting. For each j ,

$$H^{p-1}f_j g = f_j^p \left(\prod_{i \neq j} f_i^{p-1} \right) g,$$

so ϕ preserves (f_j) . It therefore preserves the ideal generated by any subset of the maximal permanents. The induced quotient splitting proves reducedness: if a^p is zero in the quotient, applying the splitting gives $a = 0$, and iteration eliminates every nilpotent. The same argument works after arbitrary field extension, so these quotients are geometrically reduced.

4.2 Homogeneous specialization

The next elementary fact will transfer both complete-intersection and critical-locus bounds from one characteristic to zero.

Lemma 4.1. *Let $J \subseteq \mathbb{Z}[x_1, \dots, x_N]$ be generated by homogeneous polynomials, and denote their generated ideals over $\mathbb{Q}$ and $\mathbb{F}_p$ by $J_{\mathbb{Q}}$ and J_p . In every degree d ,*

$$\dim_{\mathbb{Q}}(\mathbb{Q}[x]/J_{\mathbb{Q}})_d \leq \dim_{\mathbb{F}_p}(\mathbb{F}_p[x]/J_p)_d.$$

In particular, the Krull dimension of the characteristic-zero quotient is at most that of the characteristic- p quotient.

Proof. For a fixed d , form the integer matrix whose columns are the coefficient vectors of all products of a generator of J with a monomial of complementary degree. Its images over the two fields are precisely the degree- d parts of the generated ideals. Its rank over $\mathbb{Q}$ is at least its rank modulo p , proving the Hilbert-function inequality. The dimension assertion follows from the eventual polynomial growth of the cumulative Hilbert function. This also handles zero-dimensional graded quotients. $\square$

Proposition 4.2. *Let $f_1, \dots, f_s$ be positive-degree homogeneous polynomials over $\mathbb{Z}$. Suppose their reductions modulo p form a regular sequence and generate a radical ideal. Then their characteristic-zero ideal is a geometrically reduced complete intersection of height s .*

Proof. Write N for the number of variables and $d_i = \deg f_i$. Lemma 4.1 bounds the generic-fiber dimension by $N - s$. The height theorem gives the reverse inequality, so the characteristic-zero generators form a regular sequence. Both quotients have Hilbert series $\prod_i (1 - z^{d_i}) / (1 - z)^N$.

Set $V = \mathbb{Z}_{(p)}$ and $B = V[x]/(f_1, \dots, f_s)$. Each graded piece B_d is a finite V -module. Its rank is the dimension of the corresponding generic-fiber piece; the dimension of B_d/pB_d is that of the special-fiber piece. The equality of the Hilbert series makes these numbers equal. The structure theorem for finite modules over the discrete valuation ring V therefore shows that each B_d is free. In particular, B is torsion-free.

Suppose $B \otimes_V \mathbb{Q}$ had a nonzero homogeneous nilpotent. Choose a representative in some B_d by clearing a power of p , then divide out its maximal common power of p in a V -basis. The resulting element $a \in B_d$ has nonzero reduction modulo p . A power of a vanishes in the generic fiber, and torsion-freeness of the appropriate graded piece makes that power zero already in B . This contradicts reducedness of B/pB . The nilradical of a graded ring is homogeneous, so the generic fiber is reduced. A reduced finite-type algebra over the perfect field $\mathbb{Q}$ is geometrically reduced. Extension to any characteristic-zero field finishes the proof. $\square$

Apply Proposition 4.2 to the full set of maximal permanents, or to any subset, using the splitting and height arguments above. This proves the complete-intersection and reducedness assertions of Theorem 1.1, including (1.1).

5 Critical loci of row expansions

5.1 The full permanent

Append a row $u = (u_1, \dots, u_p)$ to A . Expansion along that row gives

$$\text{per}_p(A, u) = \sum_{j=1}^p u_j f_j(A). \quad (5.1)$$

Over an algebraically closed field of characteristic p , its critical equations are

$$f_1(A) = \dots = f_p(A) = 0, \quad J_f(A)^t u = 0. \quad (5.2)$$

For the rank- r stratum of the base zero set, Proposition 3.4 gives dimension at most $N - (2p - r)$. The fiber in the u -variables has dimension $p - r$. Hence the corresponding part of the critical locus has dimension at most

$$N - (2p - r) + (p - r) = p^2 - 2p.$$

There are finitely many ranks, so this bounds the entire critical locus. The subspace consisting of matrices with two prescribed zero rows has dimension $p^2 - 2p$ and satisfies all the cofactor equations. Thus the codimension is exactly $2p$.

The critical equations are homogeneous with integer coefficients. Lemma 4.1 gives the same lower bound on codimension in characteristic zero, and the two-zero-row subspace again gives equality. This argument covers $p = 2$ as well: the critical locus of $x_{11}x_{22} + x_{12}x_{21}$ is the origin.

The permanent is a nonzero squarefree polynomial over every field. Indeed, its degree in each variable is one; a repeated nonconstant factor would depend on some variable and give degree at least two in that variable. The hypersurface Jacobian criterion therefore identifies its singular set with its gradient zero set after extension to an algebraic closure. Equation (5.1) places the gradient zero set on the hypersurface even when the characteristic divides the degree. This completes the proof of (1.2) and Theorem 1.1.

5.2 Partial row expansions

Fix $S \subseteq \{1, \dots, p\}$ with $|S| = s \geq 1$ and retain only the variables u_j with $j \in S$. The critical equations of g_S are $f_j(A) = 0$ for $j \in S$ and $J_{f_S}(A)^t u = 0$. The subset version of Proposition 3.4 gives, at Jacobian rank r , base dimension at most $N - (2s - r)$ and fiber dimension $s - r$. Thus

$$\dim \text{Crit}(g_S) \leq N - s,$$

or codimension at least $2s$ in $\mathbb{A}^{N+s}$. Conversely the locus $u = 0$, $f_j(A) = 0$ for $j \in S$, has dimension $N - s$, by the height statement. It lies in the critical locus. Equality follows in characteristic p .

Lemma 4.1 transfers the lower codimension bound to characteristic zero. Proposition 4.2 gives height s for the subset ideal in that characteristic, so the same $u = 0$ locus attains equality there. The strength assertion will follow in Section 7.

6 Propagation to arbitrary matrix sizes

We require a dimension comparison for polynomial equations whose highest-degree parts are known.

Lemma 6.1. *Let $h_1, \dots, h_t$ be nonconstant polynomials over an algebraically closed field, and let h_i^{top} be their highest total-degree homogeneous parts. Then*

$$\dim V(h_1, \dots, h_t) \leq \dim V(h_1^{\text{top}}, \dots, h_t^{\text{top}}). \quad (6.1)$$

Proof. Let $J = (h_1, \dots, h_t)$, and give $k[x]/J$ the total-degree filtration. The associated graded quotient is $k[x]/\text{in}_{\deg}(J)$, where the initial ideal is generated by the highest-degree parts of all elements of J . The filtered algebra and its associated graded algebra have the same Krull dimension, as follows from the growth of the degree filtration. Since

$$(h_1^{\text{top}}, \dots, h_t^{\text{top}}) \subseteq \text{in}_{\deg}(J),$$

the dimension inequality follows. The initial ideal may strictly contain the ideal of the displayed leading parts; the inclusion in this direction is all that is needed. $\square$

Proposition 6.2. *Over any field, the geometric codimensions satisfy*

$$c_k(r, n) \geq c_k(r-1, n-1) \quad (2 \leq r \leq n), \quad (6.2)$$

$$s_k(n) \geq s_k(n-1) \quad (n \geq 3). \quad (6.3)$$

Proof. Work over an algebraic closure. Cover the nonzero matrices by the finitely many open subsets on which a specified entry is nonzero. By row and column permutations, it suffices to work with the entry in position $(1, 1)$. Fix the first row and the first column as parameters, including $a = x_{11} \neq 0$, and denote the free lower block by Y .

First consider an $r \times n$ matrix. For each set J of $r-1$ columns of Y , the maximal permanent using these columns and column one has the form

$$a \text{ per}(Y_J) + \text{terms of degree at most } r-2 \text{ in } Y. \quad (6.4)$$

To obtain $r-1$ free entries, all bottom rows must use free columns, so the top row must use column one. This proves the leading-term assertion. On every fixed-border fiber, Lemma 6.1 bounds the dimension of the common zero set of these selected equations by

$$(r-1)(n-1) - c_k(r-1, n-1).$$

The full permanental zero set in the fiber is a subset of this zero set. There are $r+n-1$ border parameters. The dimension of the full zero set on the open chart is therefore at most $rn - c_k(r-1, n-1)$. This proves (6.2) on each nonzero chart. The omitted origin has dimension zero and does not enlarge this bound.

For square critical loci, write the fixed-border matrix as

$$X = \begin{pmatrix} a & b^t \\ c & Y \end{pmatrix}, \quad a \neq 0,$$

where Y has row and column indices $1, \dots, n-1$. Among the gradient equations choose $C_{ij} = \text{per}(X_{\widehat{i+1}, \widehat{j+1}})$ for all $1 \leq i, j \leq n-1$. Expanding according to the entries selected in the first row and column gives

$$C_{ij} = a \text{ per}(Y_{\widehat{i}, \widehat{j}}) + \sum_{\substack{u \neq i \\ v \neq j}} c_u b_v \text{ per}(Y_{\widehat{\{i, u\}}, \widehat{\{j, v\}}}). \quad (6.5)$$

Here a hat indicates deletion of the specified rows or columns, and the permanent of the empty matrix is one. Each matching in C_{ij} has $n - 1$ entries. If it uses a , the remaining $n - 2$ entries lie in Y . If it avoids a , it must use two distinct border entries, one in the first row and one in the first column; only $n - 3$ entries then lie in Y . Since the border is fixed, the summands in the second term of (6.5) have degree at most $n - 3$ in Y . Consequently

$$C_{ij}^{\text{top}} = a \text{ per}(Y_{i,j}^{\wedge}).$$

These are a times all the gradient coordinates of $\text{per}_{n-1}(Y)$. Lemma 6.1 bounds the common zero set of the selected equations in each fixed-border fiber by $(n - 1)^2 - s_k(n - 1)$. The full critical fiber is a subset of that zero set. Adding the $2n - 1$ border parameters gives $n^2 - s_k(n - 1)$. The nonzero charts and the origin cover the critical locus, proving (6.3). $\square$

Proof of Theorem 1.2. Iterate (6.3) from n down to q and apply Theorem 1.1 in characteristic zero. This gives $s_k(n) \geq 2q$. Two prescribed zero rows give the upper bound $s_k(n) \leq 2n$ for every $n \geq 2$.

Iterating (6.2) gives $c_k(n - 1, n) \geq c_k(q - 1, q) = q$. There are n maximal permanent generators, so the height theorem gives $c_k(n - 1, n) \leq n$. Finally, the prime number theorem implies $q/n \rightarrow 1$. The two limits follow by squeezing between the displayed bounds. $\square$

Remark 6.3. The same propagation proves $s_k(n) \geq 2p$ for every $n \geq p$ in a fixed positive characteristic p . It does not justify using the largest prime below n in that fixed characteristic: the coefficient proof at order q uses characteristic q .

7 Strength and scope

For clarity, all strength statements use the following convention.

Definition 7.1. For a nonzero homogeneous polynomial f of degree at least two, $\text{str}(f)$ is the least t such that

$$f = \sum_{i=1}^t a_i b_i,$$

where a_i, b_i are positive-degree homogeneous polynomials and $\deg a_i + \deg b_i = \deg f$ for each i .

Lemma 7.2. *Over any field,*

$$\text{codim Crit}(f) \leq 2 \text{str}(f).$$

Proof. For a decomposition into t products, every first derivative vanishes wherever all $2t$ factors vanish. Their common zero set is nonempty, since the factors are homogeneous of positive degree, and has codimension at most $2t$ by the height theorem. The critical locus contains it, proving the claim. $\square$

Expansion along one row expresses per_n as a sum of n products, so its strength is at most n . Theorem 1.2 and Lemma 7.2 give the lower bound q in characteristic zero. At prime order, Theorem 1.1 gives exact strength p in characteristic p and zero. Similarly, g_S is a sum of s displayed products and has critical codimension $2s$, so its strength is s . This completes Theorem 1.3.

These results determine dimensions, rather than irreducible components. They also do not determine the exact singular codimension $2n$ at every composite size, or reducedness of the corresponding maximal-permanent ideal at every composite number of columns. Both questions remain natural extensions of the prime-size argument.

The bounds concern specific algebraic invariants. Critical codimension $2n$, by itself, does not imply a large arithmetic circuit. For example,

$$h(z) = \sum_{i=1}^n \prod_{j=1}^n z_{ij}$$

has a circuit with $O(n^2)$ arithmetic operations. Its critical locus requires at least two zero entries in each row and hence has codimension $2n$. Thus a circuit application of the permanental geometry would require an additional argument.

8 Exact finite checks

The supplementary verification program uses integer arithmetic and finite-field arithmetic. It checks the trace coefficient in the small cases $p = 2, 3$, verifies instances of the generator-change congruence in a truncated polynomial ring, and compares the row-expansion critical equations with the full cofactor equations on every 2×2 matrix over $\mathbb{F}_2$ and every 3×3 matrix over $\mathbb{F}_3$. A separate finite-field calculation checks the cyclic fixed-point counts for additional primes. The program also compares complete monomial expansions on both sides of (6.5) for every selected cofactor at sizes $3 \leq n \leq 7$, including their degrees in the free block. These checks are reproducible controls on the formulas. The all-prime assertions, the geometric rank-stratum estimate, and the specialization and border arguments are established by the proofs above.

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References

- [1] A. Boralevi, E. Carlini, M. Michałek, and E. Ventura, *On the codimension of permanental varieties*, *Advances in Mathematics* **461** (2025), article 110079. doi:10.1016/j.aim.2024.110079.
- [2] T. Chau, *The F -singularities of algebras defined by permanents*, *Collectanea Mathematica* **77** (2026), 843–859. doi:10.1007/s13348-025-00494-8.
- [3] A. Knutson, *Frobenius splitting, point-counting, and degeneration*, preprint (2009). arXiv:0911.4941v1.